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* One Way ANOVA

$$Y_{ij} = \mu_i + \varepsilon_{ij} \quad , j \in [n_i] , \; i \in [r]$$
$$= \mu + \alpha_i + \varepsilon_{ij} \quad \text{s.t.} \sum_{i=1}^{r} n_i \alpha_i = 0$$

→ Estimate: $(\mu, \underset{\sim}{\alpha})$ using OLS

- Assumptions: $E(\varepsilon_{ij}) = 0$
  ε_{ij} are independent

We want to minimize:

$$G(\mu, \underset{\sim}{\alpha}) = \sum_{i=1}^{r} \sum_{j=1}^{n_i} (Y_{ij} - \mu - \alpha_i)^2$$

$$\underset{\sim}{Y} = \begin{pmatrix} \mu \\ \mu \\ \vdots \\ \mu \\ \mu \end{pmatrix} + \begin{pmatrix} \mathbb{1}_{n_1} & & & & \\ & \mathbb{1}_{n_2} & & 0 & \\ & & \ddots & & \\ & 0 & & \mathbb{1}_{n_{r-1}} & \\ & & & & \mathbb{1}_{n_r} \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_{r-1} \\ \alpha_r \end{pmatrix} + \underset{\sim}{\varepsilon}$$

$$= \mu I_n + X \underset{\sim}{\alpha} + \underset{\sim}{\varepsilon} \qquad (n = n_1 + n_2 + \dots + n_r)$$

$\therefore G(\mu, \underset{\sim}{\alpha}) = \|\underset{\sim}{Y} - \mu I_n - X \underset{\sim}{\alpha}\|_2^2$ → It's a convex func^n.

Also, for this function, local minima is global minima.

→ Define $\widetilde{X} = (I - P_{\mathbb{1}})X$, $P_{\mathbb{1}} = \frac{1}{n} \mathbb{1} \mathbb{1}^T$

Constraint: $\sum n_i \alpha_i = 0 \iff \mathbb{1}^T X \underset{\sim}{\alpha} = 0$

$$\therefore \widetilde{X} \underset{\sim}{\alpha} = X \underset{\sim}{\alpha}$$

$P_X = P_{\mathbb{1}} + P_{\widetilde{X}}$ (We know)

$\therefore G(\mu, \underset{\sim}{\alpha}) = \|Y - \mu \mathbb{1}_n - \tilde{X} \underset{\sim}{\alpha}\|_2^2$

$= \|Y - P_X Y + P_X Y - \mu \mathbb{1}_n - \tilde{X} \underset{\sim}{\alpha}\|^2$

$(I - P_X)Y$ is in the orthogonal space of $\mathcal{C}(X)$

$P_X Y - \mu \mathbb{1}_n - \tilde{X} \underset{\sim}{\alpha} \in \mathcal{C}(X)$.

$\therefore G(\mu, \underset{\sim}{\alpha}) = \underbrace{\|(I - P_X)Y\|_2^2}_{T_1} + \|P_X Y - \mu \mathbb{1}_n - \tilde{X} \underset{\sim}{\alpha}\|_2^2$

(Pythagoras theorem)

$= T_1 + \|P_{\mathbb{1}} Y - \mu \mathbb{1}_n + (P_X - P_{\mathbb{1}})Y - \tilde{X} \underset{\sim}{\alpha}\|_2^2$

$P_{\mathbb{1}} Y - \mu \mathbb{1}_n$ is in $\mathcal{C}(\mathbb{1}_n)$

$P_X - P_{\mathbb{1}} = P_{\tilde{X}}$, $P_{\tilde{X}} Y - \tilde{X} \underset{\sim}{\alpha} \in \mathcal{C}(\tilde{X})$

$\because \tilde{X} = (I - P_{\mathbb{1}})X$, $P_{\mathbb{1}} Y - \mu \mathbb{1}_n \perp P_{\tilde{X}} Y - \tilde{X} \underset{\sim}{\alpha}$

$\therefore G(\mu, \underset{\sim}{\alpha}) = T_1 + \|P_{\mathbb{1}} Y - \mu \mathbb{1}_n\|_2^2 + \|P_{\tilde{X}} Y - \tilde{X} \underset{\sim}{\alpha}\|_2^2$

$= \underbrace{\|Y - P_X Y\|_2^2}_{\text{constant}} + \underbrace{\|P_{\mathbb{1}} Y - \mu \mathbb{1}_n\|_2^2}_{\text{has } \mu} + \underbrace{\|P_{\tilde{X}} Y - X \underset{\sim}{\alpha}\|_2^2}_{\text{has } \underset{\sim}{\alpha}}$

Lets minimize one by one:

$\|P_{\mathbb{1}} Y - \mu \mathbb{1}_n\|_2^2 = \|(\bar{Y}_{..} - \mu)\mathbb{1}_n\|^2$

$\therefore \hat{\mu} = \bar{Y}_{..} = \frac{1}{n} \sum_{i=1}^{r} \sum_{j=1}^{n_i} Y_{ij}$ is the OLS est. of μ.

For the other term, we need $P_{\tilde{X}} Y - X \underset{\sim}{\alpha} = 0$

$\Rightarrow P_{\tilde{X}} Y = X \underset{\sim}{\alpha}$

We know, $Ax = b$ has a sol^n when $b \in \mathcal{C}(A)$.

In our case, $P_{\tilde{X}} Y \in \mathcal{C}(X)$, so were good to go

$\hat{\underset{\sim}{\alpha}} = (X^T X)^{-1} X^T P_{\tilde{X}} Y$

$\therefore \hat{\mu}_{OLS} = \bar{Y}_{..}$

$\hat{\underset{\sim}{\alpha}}_{OLS} = (X^T X)^{-1} X^T P_{\tilde{X}} Y$

WSS → within sum of squares

BG → between groups variation

- Simplifying $\hat{\alpha}_{OLS}$:

$$(X^T X)^{-1} = \operatorname{diag}\left(\frac{1}{n_1}, \frac{1}{n_2}, \dots, \frac{1}{n_r}\right)$$

$$X^T P_{\tilde{X}} Y = X^T P_X Y - X^T P_{\mathbb{1}} Y$$

$$= X^T Y - \bar{Y}_{..} X^T \mathbb{1}_n \quad (X^T P_X = (P_X X)^T = X^T)$$

$$= \begin{pmatrix} \sum_{j=1}^{n_1} Y_{1j} \\ \vdots \\ \sum_{j=1}^{n_r} Y_{rj} \end{pmatrix} - \begin{pmatrix} n_1 \bar{Y}_{..} \\ \vdots \\ n_r \bar{Y}_{..} \end{pmatrix}$$

$$\therefore \quad \hat{\underset{\sim}{\alpha}}_{OLS} = \begin{pmatrix} \bar{Y}_{1.} - \bar{Y}_{..} \\ \vdots \\ \bar{Y}_{r.} - \bar{Y}_{..} \end{pmatrix}$$

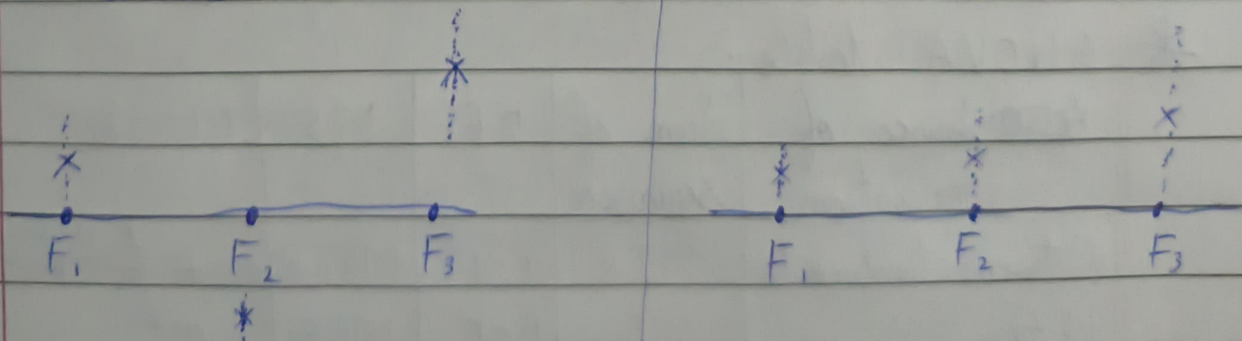
- Also, its trivial to check that $\hat{\alpha}_{OLS}$ follows the constraint: $\sum_{i=1}^{r} n_i \alpha_i = 0$

• Model: $\underset{\sim}{Y} = \mu \mathbb{1}_n + X \underset{\sim}{\alpha} + \underset{\sim}{\varepsilon}$

OLS estimates:

$$\boxed{\hat{\mu}_{OLS} = \bar{Y}_{..} \; ; \; \hat{\alpha}_{OLS} = \begin{pmatrix} \bar{Y}_{1.} - \bar{Y}_{..} \\ \vdots \\ \bar{Y}_{r.} - \bar{Y}_{..} \end{pmatrix}}$$

Scene 1 | Scene 2



WSS << BG $\quad \frac{BG}{WSS}$ is large.

$\frac{BG}{WSS}$ is small

Variance decomposition:

$$(I - P_{\mathbb{1}})Y = (I - P_X)Y + (P_X - P_{\mathbb{1}})Y = (I - P_X)Y + P_{\tilde{X}}Y$$

$$\perp$$

$$\Rightarrow \cancel{(I-P_{\mathbb{1}})} \|(I - P_{\mathbb{1}})Y\|_2^2 = \|(I - P_X)Y\|_2^2 + \|P_{\tilde{X}}Y\|_2^2$$

$$\Rightarrow Y^T(I - P_{\mathbb{1}})Y = Y^T(I - P_X)Y + Y^T P_{\tilde{X}} Y$$

$$\Rightarrow \quad SST_o = SSE + SSR$$

$$SSE = \sum_{i=1}^{r}\sum_{j=1}^{n_i}(Y_{ij} - \overline{Y}_{i\cdot})^2 \rightarrow WG \text{ (within group)}$$

$$SSR = \sum_{i=1}^{r} n_i(\overline{Y}_{i\cdot} - \overline{Y}_{\cdot\cdot})^2 \rightarrow BG$$

\# Assuming $\varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma^2)$, deriving the sampling dist^n of $\dfrac{SSR}{SSE}$.

i) Show that $\dfrac{SSR/r-1}{SSE/n-r} \sim F_{r-1, n-r}$ Under H_0.

Note: $MSR = SSR/r-1$; $MSE = SSE/n-r$

$H_0: \alpha_1 = \alpha_2 = \ldots = \alpha_r$

ii) Under H_1, what'll be dist^n of the same ratio?

* ANOVA table:

~~ANO~~ Source of variation	Sum of squares	D.F.	MS	E(MS)
Between treatments	SSTR	$r-1$	MSTR	$\sigma^2 + \sum_{i=1}^{r} \frac{n_i(\mu_i - \mu)^2}{r-1}$
Within "	SSE	$n-r$	MSE	σ^2
Total	SSTO	$n-1$		

F-ratio = $\frac{MSTR}{MSE}$. Finally conclude whether H_0 is rejected.